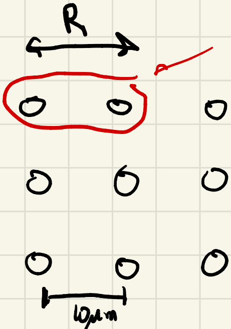
What about two qubit gates?

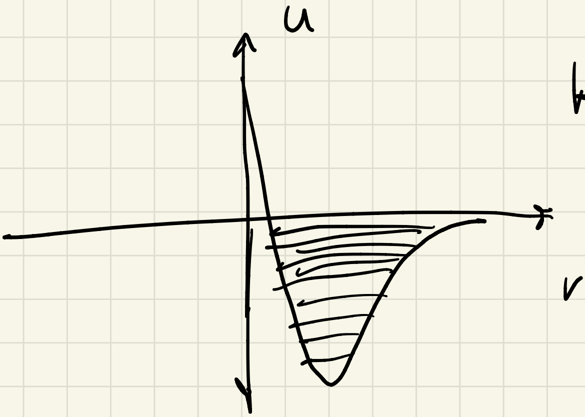


want to entangle adjacent qubits.

Use the Rydberg states (high quantum ~~#~~ S states)

~100S

Hydrogenic potential.



Spatially are large → classical argument:

$$F = \frac{e^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r}$$

$$\downarrow \qquad \downarrow$$

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{m n^2 \hbar^2}{m^2 r^3} \; ; \qquad \boxed{r = \frac{4\pi\epsilon_0 n^2 \hbar^2}{e^2 m}}$$

momentum: $mvr = n\hbar$

$$v = \frac{n\hbar}{mr}$$

$\frac{C}{m^2}$ $a_0 = 5.29\times10^{-11}\,m$ $e = 1.6\times10^{-19}\,C$

Dipole - Dipole interaction. $|r\rangle \leftrightarrow |r\rangle$

$$V_{rr} \approx \frac{(n^2 e a_0)^2}{R^3} \sim 250\,MHz$$

Compare to atomic linewidth $\sim 6\,MHz$

- large electronic dipole-dipole interaction between the atoms.
- Scales as $\frac{1}{R^3}$ up to $\sim 10\,\mu m$
- Compared to the interaction shift between

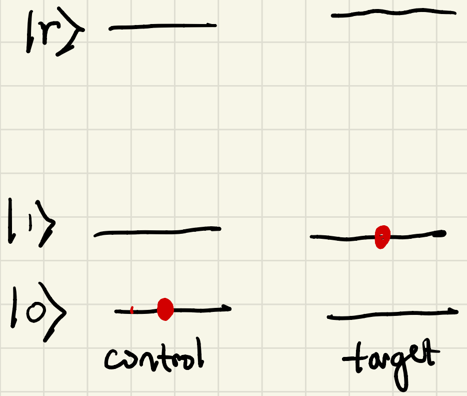
2 atoms; this is 10^{12} Larger than interaction between ground state atoms,

* Issues: finite lifetime of Rydberg States $(\sim 100\,\mu s)$ (worse for higher n numbers)

$$\tau \sim \frac{1}{\frac{1}{\tau_0 n^3} + \frac{1}{\tau_{BB}}} \longleftarrow$$

- black-body.
- τ_0 spontaneous.

Gate concept:


$|r\rangle$
$|1\rangle$
$|0\rangle$
control
target

- Possible configurations:

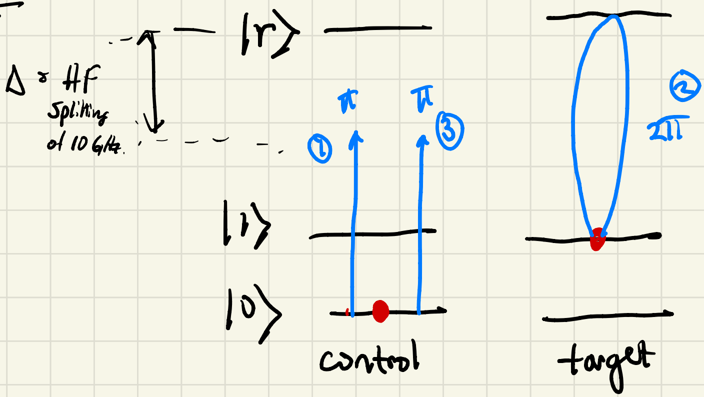
  $|01\rangle$; $|10\rangle$; $|00\rangle$; $|11\rangle$.

- Now apply pulses that will be resonant with Rydberg transitions $|1\rangle \rightarrow |r\rangle$

- Transition to the Rydberg state and back again (2π pulse) results in a $e^{i\pi}$ phase shift:

  i.e. $|01\rangle \rightarrow -|01\rangle$

Case I. $|01\rangle$; or $|10\rangle$

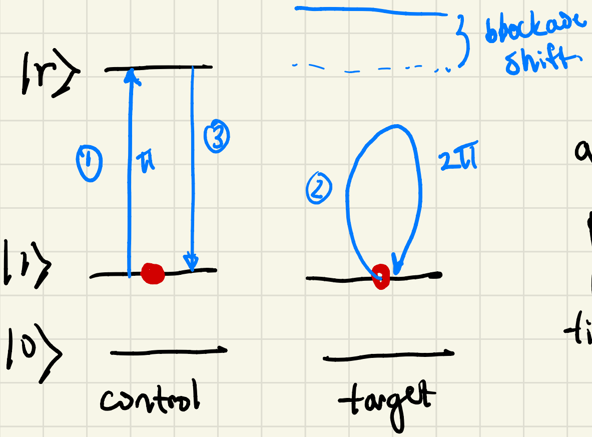

Δ = HF splitting of 10 GHz
|r⟩
|1⟩
|0⟩
①
π
π
③
②
2π
control
target

in this case, $|01\rangle$ state picks up $\Delta\phi_t = \pi$, $\Delta\phi_c = 0$

and similarly for $|10\rangle$

($\Delta\phi_t = 0$, $\Delta\phi_c = \pi$)

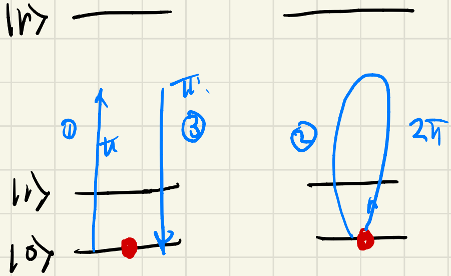
Case II


blockade shift
|r⟩
|1⟩
|0⟩
①
π
③
②
2π
control
target

again, a relative phase of π is picked up, but this time by the control.

$$\Delta\phi_t = 0.$$
$$\Delta\phi_c = \pi$$

Case III


|r⟩
|1⟩
|0⟩
①
π
π
③
②
2π

in this case nothing happens.

$$\Delta\phi_c = \Delta\phi_t = 0.$$

This implements a C-Phase gate: (CZ gate)

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

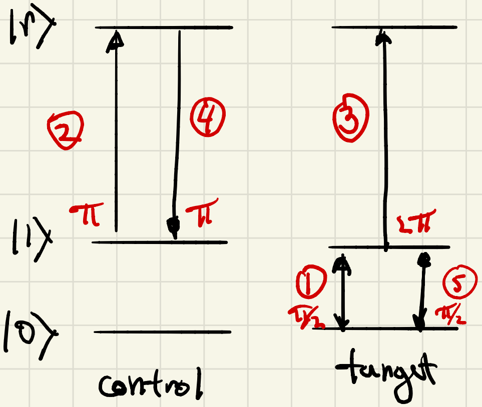
This is CZ as shown in Rydberg literature.

Typical CZ gate:

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

→ can obtain through phase shift and flip of basis.

How do you obtain a CNOT gate?

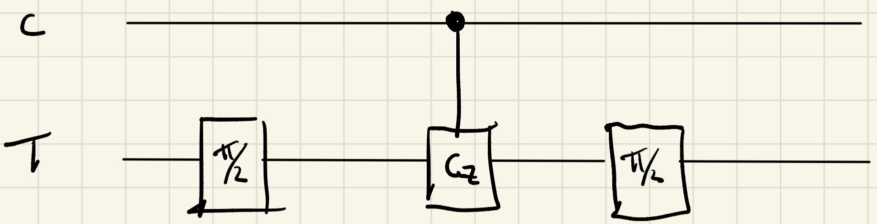

|r⟩
|1⟩
|0⟩
②
④
③
①
⑤
π
π
2π
π/2
π/2
control
target

Apply a π/2 pulse to the target state before and after the CZ gate.

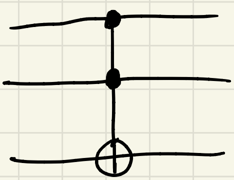
Swapping target + control, we get the following for CZ

$$-1 \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

The circuit is then: (after the swap and ignore the global phase)



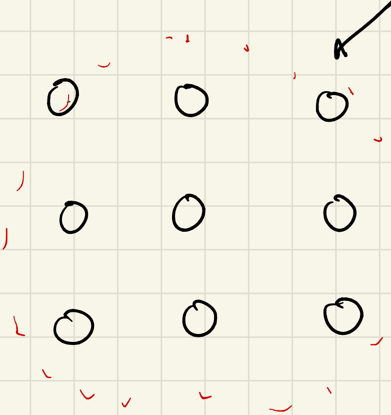
Multi qubit gate → Toffoli



- This gate will only flip the target if both control qubits are in state $|1\rangle$

- To implement this gate we can proceed similarly, first by implementing a CCZ gate:

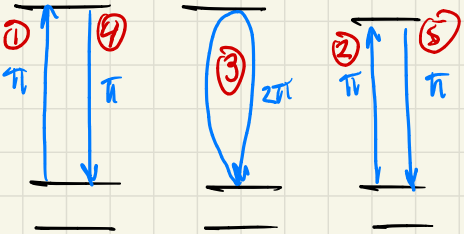
→ Why does this work?



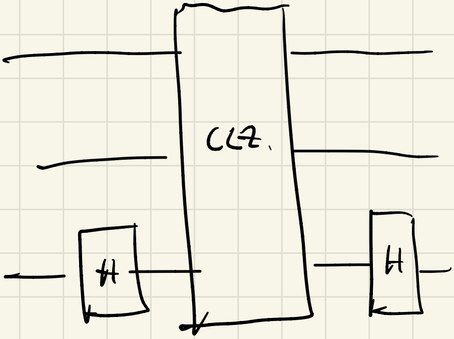
blockade radius can effects multiple neighbouring atoms

- CCZ gate
  - Similar to the CZ gate we should expect a scheme that realises $e^{i\pi}$ phase shift for everything except $|000\rangle$.

Pulse sequence: (CCZ)



Toffoli:



Note that implementing this w/ 2-qubit gates (i.e. CNOT) requires 6 CNOTs, 2 H gates and 7 T gates.

- This requires 30 laser pulses, vs. 5 for the Toffoli.

⟶ native implementation could be a significant advantage.

What about gate fidelities?

Aim for similar 10^{-4} fidelities:

$F \sim 0.9999$

Errors related to lifetime and Blockade Strength:

$$E_{min} = \frac{3(7\pi)^{2/3}}{8} \cdot \frac{1}{(B\tau)^{2/3}}$$

Best case scenario: 2×10^{-5} ✓

(may be worse for 2 qubit superposition inputs however)

✓

numerical simulations $\rightarrow$ $E_{min} \sim 10^{-3}$.

There is potential for improved schemes.

Experimentally: $\longrightarrow$ ~ 94.1% (2018)

Many additional issues not worth covering here

- inhomogeneous fields
- additional states (hyperfine states)

2023 Results (see additional materials)

Two-qubit fidelity of 99.6% (above E.C. threshold)

Outlook:

- Why still be interested in neutral atoms?
  - **Scaling!** ~ 150 adressible qubits demonstrated.

    could imagine 200 x 200 arrays are within the range of feasibility w/ current technology.

    ⟶ 40,000 qubit array ⟶ hard to imagine this soon w/ μ-wave qubits.

- Rydbergs are also useful for Quantum Simulation.

- Estimate that 10 k physical qubits achievable in next couple of years.